CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2017 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Metadata and Data Dictionary.

Definition: Metadata is "data about data" describing database structures, constraints, and data types. A **Data Dictionary** is the repository that permanently stores this metadata.

Q2 (3M) Define Tuple, Relation, Attribute, and Domain.

Definitions:

- **Relation:** A two-dimensional table of records.
- **Tuple:** A single row in a relation representing an individual entity instance.
- **Attribute:** A named column describing a specific property.
- **Domain:** The set of allowable atomic values for an attribute.

Q3 (3M) What are Cardinality and Participation Constraints in ER Modeling?

Definitions:

- **Cardinality Ratio:** Specifies the maximum number of relationship instances that an entity can participate in (1:1, 1:N, M:N).
- **Participation Constraint:** Specifies whether existence of an entity depends on its being related to another entity: *Total* (double line) or *Partial* (single line).

Q4 (3M) Differentiate between DDL and DML.

DDL: Defines database schemas and tables (`CREATE` , `ALTER` , `DROP`).
DML: Manipulates and retrieves data instances (`INSERT` , `UPDATE` , `DELETE` , `SELECT`).

Q5 (3M) What is a View in SQL?

Definition: A **View** is a virtual table whose contents are defined by an underlying SQL query, providing security and abstraction without replicating stored data.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain 1NF and 2NF with examples.

1NF: Eliminates repeating groups and non-atomic values.
2NF: In 1NF and contains no partial functional dependencies (all non-key attributes depend fully on the primary key).

Q7 (7.5M) **What is Query Optimization? Explain heuristic optimization.**

Explanation: Query optimization is the process of choosing the most efficient execution plan for evaluating a query. Heuristic optimization applies algebraic rewrite rules:

- Perform `Selection (σ)` as early as possible (reduces intermediate relation sizes).
- Perform `Projection (π)` early to eliminate unneeded attributes.
- Execute the most restrictive selections before performing Joins.

Q8 (7.5M) **Explain Log-Based Recovery with WAL (Write-Ahead Logging) and Checkpoints.**

Log-Based Recovery: Records all database modifications in an append-only log file before writing changes to disk (Write-Ahead Logging). During crash recovery, log records are used to UNDO incomplete transactions and REDO committed ones. Checkpoints reduce recovery overhead by flushing dirty buffers periodically.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the Three-Level ANSI/SPARC Database Architecture with detailed diagram.**

Architecture: External Schema (Views) → Conceptual Schema (Logical tables) → Internal Schema (Physical disk blocks and access paths). Ensures physical and logical data independence.

Q10 (15M) **Given tables VENDOR(VCode, Name, Address, Phone) and PRODUCT(PCode, PName, Price, VCode). Write SQL queries for:**
(i) Display all vendors from 'Delhi'.
(ii) Display products with price greater than 500.
(iii) Display vendor name who supplies product 'Laptop'.
(iv) Count total products supplied by each vendor.

SQL Solutions:

```
-- (i) Vendors from Delhi
SELECT * FROM VENDOR WHERE Address LIKE '%Delhi%';

-- (ii) Products with price > 500
SELECT PCode, PName, Price FROM PRODUCT WHERE Price > 500;

-- (iii) Vendor supplying 'Laptop'
SELECT V.Name
FROM VENDOR V
JOIN PRODUCT P ON V.VCode = P.VCode
WHERE P.PName = 'Laptop';

-- (iv) Count total products per vendor
SELECT V.VCode, V.Name, COUNT(P.PCode) AS TotalProducts
FROM VENDOR V
LEFT JOIN PRODUCT P ON V.VCode = P.VCode
GROUP BY V.VCode, V.Name;
```

Q11 (15M) **Explain Relational Algebra vs Relational Calculus.**

Comparison: Relational Algebra is procedural (specifies operations step-by-step), while Relational Calculus (Tuple & Domain) is non-procedural/declarative (expresses what is required via mathematical predicate logic). Codd's Theorem proves they are equivalent in expressive power.

Q12 (15M) **Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.**

BCNF: Stricter form of 3NF where every determinant in a non-trivial functional dependency must be a super key. Eliminates anomalies that 3NF allows when there are overlapping composite candidate keys.

Q13 (15M) **Given tables: PLAY(PlayCode, PlayName, DirNum), DIRECTOR(DirNum, DirName, DirDOB). Write SQL queries to find director names and plays directed.**

SQL Solutions:

```
-- Director of play 'Hamlet'
SELECT D.DirName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
WHERE P.PlayName = 'Hamlet';

-- All plays directed by each director
SELECT D.DirName, P.PlayName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
ORDER BY D.DirName;
```
